

TermPlanMT

Graph-Grounded Terminology Planning for French → English Regulatory Medical Document Translation

Machine Translation Project Proposal · Darragh Kerins & [Partner] · UPV/EHU

Introduction

Neural MT produces fluent output but does not enforce consistent terminology. In regulatory medical documents the same term must appear identically across hundreds of segments, and mistranslation can have direct patient-safety consequences. Three failure modes are common:

- **Term drift** — the same source term is translated differently across a document.
- **Concept flattening** — a specific term is replaced by an over-general hypernym.
- **Coverage gaps** — domain-specific terms are omitted or rendered inconsistently.

The EMA publishes Summaries of Product Characteristics (SmPCs) for every approved drug across all 24 EU languages, each with a certified human English translation. MedDRA provides a validated five-level ontology covering adverse events, indications, and drug classes:

Level	Tier	Example
L1	System Organ Class (SOC)	Respiratory disorders
L2	High Level Group Term (HLGT)	Pulmonary conditions
L3	High Level Term (HLT)	Pneumonitis
L4	Preferred Term (PT)	Pembrolizumab-related pneumonitis
L5	Lowest Level Term (LLT)	Immune-mediated pneumonitis

TermPlanMT connects these two resources: it reads a source French SmPC, extracts medical terms with a French biomedical NER model, grounds each term in a persistent Neo4j concept graph built from MedDRA, and enforces that each translated English term lands at the same MedDRA hierarchy level as its French source. The Neo4j graph is the interpretable layer of the pipeline — every terminology decision is traceable to a specific node, its MedDRA level, and its neighbours.

Scope

French–English Keytruda (pembrolizumab) SmPC, Section 4.8 (Undesirable Effects) — 10 pages, ≈4,400 words, ≈5,700 tokens. This section exhibits all three failure modes at scale and contains the densest domain-specific terminology in the document.

Data access

A MedDRA non-commercial academic licence has been applied for through the MSSO under UPV/EHU. SNOMED-CT is a freely accessible fallback.

Research Questions

1. Does graph-based terminology planning (MedDRA + Neo4j) improve TCR over zero-shot NMT?
2. Does a long-context LLM with full-document awareness improve consistency without explicit ontology grounding?
3. Does injecting structured ontology context into an LLM (GraphRAG, no constraints) further improve hierarchy-level accuracy?
4. Does post-hoc reranking over the ontology graph outperform unguided GraphRAG generation?
5. Does decoding-level constraint enforcement add measurable gains over reranking alone?

Related Work

Constrained decoding. Hokamp & Liu [1] introduced grid beam search for hard lexical constraints; Post & Vilar [2] extended this with dynamic beam allocation (DBA), used in Stage 4 of TermPlanMT. Neither approach addresses *which* constraint to select when multiple ontologically related candidates exist — that is what the planning step solves.

Terminology in NMT. Dinu et al. [3] and Bergmanis & Pinnis [4] employ flat glossaries with no concept structure. TermPlanMT selects constraints from a hierarchy, not a list, and the planning step scores candidates against graph-adjacent renderings.

Long-context LLMs. Herold & Ney [5] show that extending NMT context windows degrades performance due to attention overload. By contrast, instruction-tuned LLMs such as Mistral 7B process multi-page documents naturally, motivating their use as both a standalone baseline (S2) and a GraphRAG backbone (S3–S5).

Biomedical NER. Martin et al. [6] provide CamemBERT for French; English-only biomedical NER models are not appropriate for the source language here.

MT evaluation. BLEU [7], chrF [8], and COMET [9] measure fluency but none captures hierarchy preservation — motivating the HTM metric introduced in Section 7.3.

Data

The KEYTRUDA (pembrolizumab) SmPC from the EMA product database [10] (FR source: 368 pages, ≈129,700 words; EN reference: 341 pages). Evaluation focuses on Section 4.8 — pages 15–24 of the EN reference — which exhibits all three failure modes at scale:

- **Term drift:** *pneumopathie immuno-médiée* appears in 30+ segments; zero-shot MT produces at least three distinct English renderings.
- **Concept flattening:** LLT-level terms (L5) are frequently collapsed to *pneumonitis* (L3/L4) — two specificity levels too general.
- **Coverage gaps:** multi-token adverse-event names spanning clause boundaries fall outside standard NER and are left unconstrained.

System Architecture

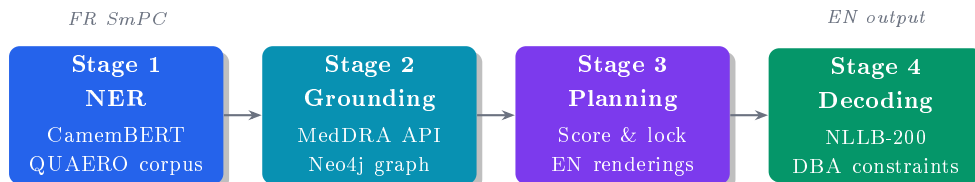


Figure 1: TermPlanMT four-stage pipeline.

Stage 1 — French Term Extraction

A CamemBERT-based NER model [6] fine-tuned on the QUAERO French Medical corpus [11] processes the source SmPC in a single pass, tagging drug names, drug classes, adverse events, anatomical terms, dosage forms, and quantities. No additional fine-tuning is required.

Stage 2 — MedDRA Grounding and Persistent Neo4j Graph

Each extracted French term is looked up in MedDRA via its REST API, returning a concept identifier, preferred English label, hierarchical level, and parent/child identifiers. These are written into a persistent Neo4j graph: MedDRA’s five levels as node types, `BROADER_THAN` relations as edges. Coverage is tracked with $CCR = |\text{grounded}| / |\text{extracted}|$.

Listing 1: Neo4j grounding — MERGE pattern preserves graph across runs.

```
from neo4j import GraphDatabase
driver = GraphDatabase.driver("bolt://localhost:7687", auth=("neo4j", "password"))

def ground_term(tx, fr_term, concept_id, en_label, level, parent_id):
    tx.run("""
```

```

MERGE (c:Concept {id: $concept_id})
SET c.fr_label = $fr_term, c.en_label = $en_label, c.level = $level
WITH c MATCH (p:Concept {id: $parent_id})
MERGE (p)-[:BROADER_THAN]->(c)
""", concept_id=concept_id, fr_term=fr_term,
    en_label=en_label, level=level, parent_id=parent_id)

```

Stage 3 — Terminology Planning

When MedDRA returns multiple valid English renderings, each candidate c is scored:

$$\text{Score}(c) = \alpha \cdot \text{Context}(c) + \beta \cdot \text{Neighbour}(c) - \gamma \cdot \text{HierarchyPenalty}(c) \quad (1)$$

- $\text{Context}(c)$: cosine similarity to the local sentence (`paraphrase-multilingual-mpnet-base-v2`).
- $\text{Neighbour}(c)$: mean similarity to the selected renderings of graph-adjacent concepts (parents, children, siblings), retrieved via Cypher.
- $\text{HierarchyPenalty}(c) = |d_{\text{MedDRA}} - d_{\text{source}}|$: penalises candidates at the wrong specificity level.

Weights $\alpha, \beta, \gamma \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ are grid-searched on the dev set. A global consistency pass locks the top-scoring rendering for each term before any segment is translated.

Stage 4 — Constrained French→English Decoding

NLLB-200 [12] (`fra_Latn`→`eng_Latn`) with Dynamic Beam Allocation [2] guarantees a fully-constrained hypothesis always reaches the final beam. Locked renderings from Stage 3 are passed as hard constraints. Beam size defaults to 5, expanded to 10 for segments with three or more simultaneous constraints.

Gold-Standard Term Annotation

A manual FR→EN term mapping is built for all unique medical terms in Section 4.8, with MedDRA levels verified against the official MedDRA browser.

French	English (gold)	L	Tier
Antinéoplasiques	Antineoplastic agents	1	SOC
Inhibiteurs de point de contrôle	Checkpoint inhibitors	2	HLGT
Inhibiteur PD-1	PD-1 inhibitor	3	HLT
Pembrolizumab	Pembrolizumab	4	PT
Pneumopathie immuno-médiée	Immune-mediated pneumonitis	5	LLT

Evaluation

Experimental Conditions

The five conditions form a clean causal ablation — each step adds exactly one capability, enabling direct isolation of each contribution.

ID	System	Graph	Enforce	Ablation question answered
S1	NLLB-200 zero-shot	No	No	Lower bound — pure NMT baseline
S2	Mistral 7B full-doc context	No	No	Does long-context LLM beat NMT?
S3	Mistral 7B + GraphRAG	Yes	No	Does structure alone help? (key)
S4	GraphRAG + reranking	Yes	Yes	Does post-hoc enforcement matter?
S5	GraphRAG + logit constraints	Yes	Yes	Is decoding-level control worth complexity?

Table 1: Experimental conditions. S3 (GraphRAG, no enforcement) is the key novel condition.

Causal ablation chain — each comparison isolates one variable

- **S1 vs S2:** does long-context LLM improve consistency over segment-level NMT?
- **S2 vs S3:** does ontology retrieval alone (GraphRAG) improve terminology accuracy?
- **S3 vs S4:** does post-hoc reranking over the graph add further gains?
- **S4 vs S5:** is decoding-level logit enforcement worth the added complexity?

S3 in detail — GraphRAG without constraints

Mistral 7B Instruct receives the source segment plus ontology context retrieved from Neo4j via Cypher: the MedDRA-preferred English rendering and its level for each source term injected directly into the prompt. No post-hoc filtering or logit manipulation is applied. This isolates the question: *does structured retrieval alone fix terminology errors, or is enforcement necessary?*

Listing 2: GraphRAG prompt pattern for S3.

```
system_prompt = """You are a medical translator. For each [TERM] in the source,
the MedDRA ontology provides the preferred English rendering and its specificity
level. Use this rendering exactly. Do not generalise."""

# Retrieve context from Neo4j
context = session.run("""
    MATCH (c:Concept {fr_label: $fr_term})
    RETURN c.en_label AS label, c.level AS level
""", fr_term=term)
```

Metrics

Metric	Measures	Tool
BLEU	n -gram fluency	sacrebleu
chrF	Character F-score	sacrebleu
COMET	Neural translation quality	unbabel-comet
TCR	Term translation consistency	Custom
HTM	Hierarchy-aware level accuracy	Custom (novel)
CCR	Concept grounding coverage	Custom

HTM — Hierarchy-Aware Terminology Metric

HTM measures how far a translated term has drifted from its source position in the MedDRA hierarchy. Each MT output term is grounded in Neo4j to recover its level $\hat{d}_i$, then compared against the source level d_i :

$$\text{HTM} = \frac{1}{N} \sum_{i=1}^N s_i, \quad s_i = \begin{cases} 1.0 & \hat{d}_i = d_i \text{ (exact level match)} \\ 0.5 & \hat{d}_i \neq d_i, \text{ same branch (wrong specificity)} \\ 0.0 & \text{different branch or ungrounded} \end{cases} \quad (2)$$

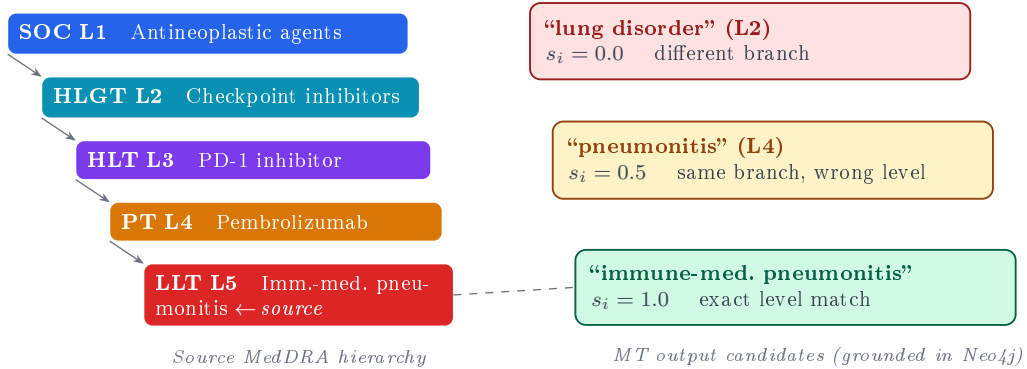


Figure 2: HTM scoring for *pneumopathie immuno-médicée* (source: LLT, L5). Each MT output is grounded in Neo4j and its MedDRA level compared against the source.

Expected Results

Success thresholds: TCR $\geq +5$ pp over S1; HTM $\geq +0.05$ over S1; BLEU/COMET within 1 point of S1.

Metric	S1 NMT	S2 LLM ctx	S3 GraphRAG	S4 Rerank	S5 Logit
TCR	~68%	~72%	~78%	~84%	~87%
HTM	~0.60	~0.63	~0.69	~0.74	~0.78
BLEU	~36.0	~37.5	~37.0	~38.5	~39.0
COMET	~0.81	~0.82	~0.82	~0.84	~0.84

S3 expected to partially close the gap to S4/S5, with largest shortfall at LLT-level (L5) terms where LLM hallucinations are most likely.

Error Analysis

Twenty randomly sampled segments per system are annotated for: **term drift** (≥ 2 renderings of the same term), **concept flattening** (over-general hypernym substituted), and **fluency loss** (grammatically correct but unnatural output). Results are broken down by SmPC section to reveal whether errors concentrate in legally critical zones.

Contributions

- **Pipeline.** French NER $\rightarrow$ MedDRA grounding via persistent Neo4j $\rightarrow$ graph-based planning $\rightarrow$ constrained NLLB-200 decoding.
- **Ablation design.** Five-condition causal ladder isolating the contribution of document context, ontology retrieval, reranking, and decoding-level enforcement.
- **HTM metric.** A novel graph-comparison metric measuring MedDRA level drift between source and translated terms.
- **Knowledge graph.** A reusable Neo4j pharmaceutical terminology graph, accumulated across all processed SmPCs.

Limitations

MedDRA coverage. Terms absent from MedDRA fall back to unconstrained translation; CCR quantifies this gap.

NER recall. Missed source terms cannot be grounded or constrained; pipeline performance is bounded by Stage 1 recall.

Fluency trade-off. Hard constraints can produce unnatural output in dense segments; beam widening mitigates but does not eliminate this.

Scope. Single document (Section 4.8), French–English only; extension to further drug classes and language pairs is future work.

References

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